Supervised Classification

Bayesian classification

Bayesian Classification

- A statistical classifier: performs *probabilistic prediction, i.e.,* predicts class membership probabilities
- Foundation: Named after *Thomas Bayes,* who proposed the *Bayes Theorem*.
- Performance:
  - It can solve problems involving both categorical and continuous valued attributes.
  - A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with that of decision tree and some neural network classifiers.

Bayesian Theorem: Basics

- Let **X** be a data sample
- Let H be a *hypothesis* that X belongs to class C
- Classification is to determine P(H|**X**), the probability that the hypothesis holds given the observed data sample **X**
  - **Example:** customer X will buy a computer given that the customer's age and income are known

Bayesian Theorem: Basics

- P(H) (*prior probability*), the initial probability
  - E.g., **X** will buy computer, regardless of age, income, …
- P(**X**) (*evidence*): probability that sample data is observed
- P(**X**|H) (*likelihood*), the probability of observing the sample **X**, given that the hypothesis holds
  - E.g., Given the hypothesis is that **X** will buy computer, then P(**X**|H) denotes the prob. that Age of X is between 31…40, having medium income

Bayesian Theorem

- Given training data **X**, *posteriori probability of a hypothesis* H, P(H|**X**), follows the Bayes theorem

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})}$$

- Let D be a training set of tuples with their associated class labels, and each tuple is represented by an n attribute vector $X = (x_1, x_2, \dots, x_n)$
- Suppose there are k classes $C_1, C_2, \dots, C_k$
- $P(C_k|X) = \frac{P(C_k,X)}{P(X)}$ (1)

$or, P(C_k|X) = \frac{P(C_k)P(X|C_k)}{P(X)}$ (2)

- Since the denominator does not depend on C_k and all the values of the features(i.e. x_i) are given, the denominator is effectively constant. So we are interested only in the numerator part of the fraction.
- The numerator is equivalent to the joint probability model $P(C_k, x_1, \dots, x_n)$

- Based on Equation 1 and 2 the numerator can be written as:

$$P(C_k)P(X|C_k) = P(C_k, X)$$

Or, $P(C_k)P(x_1, x_2, \dots, x_i, \dots x_n|C_k) = P(C_k, x_1, x_2, \dots, x_i, \dots x_n)$

Where $X = (x_1, x_2, \dots, x_i, \dots, x_n)$

- Using the chain rule for repeated applications of the definition of conditional probability we can write:

$$P(C_k, x_1, x_2, .., x_i, \dots, x_n) = P(x_1, x_2, .., x_i, \dots, x_n, C_k)$$
$$= P(x_1|x_2, .., x_i, \dots, x_n, C_k)P(x_2, .., x_i, \dots, x_n, C_k)$$
$$= P(x_1|x_2, .., x_i, \dots, x_n, C_k)P(x_2|x_3, .., x_i, \dots, x_n, C_k)P(x_3, .., x_i, \dots, x_n, C_k)$$
$$= \cdots$$
$$= P(x_1|x_2, .., x_i, \dots, x_n, C_k)P(x_2|x_3, .., x_i, \dots, x_n, C_k) \dots P(x_{i-1}|x_i, \dots, x_n, C$$
$$..P(x_{n-1}|x_n, C_k)P(x_n|C_k)P(C_k)$$

- Now the "naive" conditional independence assumptions come into play: assume that each feature x_i is conditionally independent of every other feature x_j for $j \neq i$, given the category C_k. This means that

$$P(x_i | x_{i+1}, \ldots, x_n, C_k) = P(x_i | C_k) \qquad (3)$$

- Thus, the joint model can be expressed as

$$P(C_k|x_1,\dots,x_n) \propto P(C_k,x_1,\dots,x_n)$$

$$= P(x_1|x_2,\dots,x_i,\dots,x_n,C_k)P(x_2|x_3,\dots,x_i,\dots,x_n,C_k)\dots P(x_{i-1}|x_i,\dots,x_n,C_k)\dots P(x_{n-1}|x_n,C_k)\ P(x_n|C_k)P(C_k)$$

$$= P(C_k)P(x_1|C_k)P(x_2|C_k)\ \dots.P(x_n|C_k)[from\ eqn.(3)]$$

$$= P(C_k)\prod_{i=1}^{n} P(x_i|C_k)$$

Constructing a classifier from the probability model

- The discussion so far has derived the independent feature model, that is, the naive Bayes probability model.
- The naive Bayes classifier combines this model with a decision rule.
  - One common rule is to pick the hypothesis that is most probable; this is known as the maximum a posteriori or MAP decision rule.
  - The corresponding classifier, a Bayes classifier, is the function that assigns a class label $\hat{y} = C_k$ for some k as follows:

$$\hat{y} = \underset{k \in \{1,..,K\}}{\operatorname{argmax}} P(C_k) \prod_{i=1}^{n} P(x_i | C_k)$$

Towards Naïve Bayesian Classifier

- If A_i is categorical, $P(x_i|C_k)$ is the # of tuples in C_k having value x_i for A_i divided by $|C_{k, D}|$ (# of tuples of C_k in D)

- If A_i is continous-valued, $P(x_i|C_k)$ is usually computed based on Gaussian distribution with a mean μ and standard deviation σ

- $g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

  So, $P(x_i|C_k) = g(x_i, \mu_{C_k}, \sigma_{C_k})$

Naïve Bayesian Classifier: Training Dataset

Age Group	Income	Student	Credit_rating	Class: buys_laptop
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle_aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle_aged	Low	Yes	Excellent	yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle_aged	Medium	No	Excellent	Yes
Middle_aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

Apply Naïve Bayes classifier to classify the following tuple

X = (Age_group = Youth, income = medium, student = Yes, credit_rating = fair)

Here the output will be :

X belongs to class ("buys_laptop = Yes")

or

X belongs to class ("buys_laptop = No)

- $P(C_i)$: $P(C_1)$ = P(buys_laptop = "yes") = 9/14 = 0.643
  $P(C_2)$ = P(buys_laptop = "no") = 5/14= 0.357

- Compute $P(X|C_i)$ for each class
  P(age_group = "youth" | buys_laptop = "yes") = 2/9 = 0.222
  P(age_group = "youth" | buys_laptop = "no") = 3/5 = 0.6
  P(income = "medium" | buys_laptop = "yes") = 4/9 = 0.444
  P(income = "medium" | buys_laptop = "no") = 2/5 = 0.4
  P(student = "yes" | buys_laptop = "yes) = 6/9 = 0.667
  P(student = "yes" | buys_laptop = "no") = 1/5 = 0.2
  P(credit_rating = "fair" | buys_laptop = "yes") = 6/9 = 0.667
  P(credit_rating = "fair" | buys_laptop = "no") = 2/5 = 0.4

Naïve Bayesian Classifier: An Example

$P(X|C_i)$:

P(X|buys_laptop = "yes") = 0.222 x 0.444 x 0.667 x 0.667 = 0.044

P(X|buys_laptop = "no") = 0.6 x 0.4 x 0.2 x 0.4 = 0.019

$P(X|C_i)*P(C_i)$: P(X|buys_laptop = "yes") * P(buys_laptop = "yes") = 0.028
P(X|buys_laptop = "no") * P(buys_laptop = "no") = 0.007

Therefore, X belongs to class ("buys_laptop = yes")

Avoiding the 0-Probability Problem

- Naïve Bayesian prediction requires each conditional probability be non-zero. Otherwise, the postreior probability will be zero.

$$P(X|C_i) = \prod_{k=1}^{n} P(x_k|C_i)$$

- Example. Suppose we have a dataset with 1000 tuples, income=low(0), income=medium(990), and income=high(10)
  - Use Laplacian correction (or Laplacian estimator)
  - Adding 1 to each case
    - P(income=low)=1/1003
    - P(income=medium)=991/1003
    - P(income=high)=11/1003

Naïve Bayesian Classifier

- Advantages
  - Easy to implement
  - Good results obtained in most of the cases
- Disadvantages
  - Assumption: class conditional independence, Not always valid for real life problems, since dependencies do exist among variables
    - E.g., hospitals, patient's name, age, family history, etc.
  - Dependencies among these cannot be modeled by Naïve Bayesian Classifier

Advantage of Naïve Bayes over KNN and Decision Tree

- Naïve Bayes Over KNN
  - KNN doesn't know which attributes are more important. As a result of that, While computing distance between data points (usually Euclidean distance or other generalizations of it), each attribute normally weighs the same to the total distance. This means that attributes which are not so important will have the same influence on the distance compared to more important attributes.
  - Naïve Bayes is one of the classifiers that handle missing data very well, it just excludes the attribute with missing data when computing posterior probability (i.e. probability of class given data point). With KNN, one can't do classification if there is any missing data. The reason is that, distance is undefined if one or more of attributes (which are essentially dimensions) are missing, unless these attributes are being omitted while computing distance. As a consequence, we need to rely on common solutions for missing data, e.g. imputing average values.
  - KNN needs one parameter more than Naïve Bayes. This is the number of neighbors (K). This means one need to do model selection for KNN in order to determine the best values for K.

Advantage of Naïve Bayes over KNN and Decision Tree

- KNN classifier is a supervised lazy classifier which has local heuristics. Being a lazy classifier, it is difficult to use this for prediction in real time. On the other hand, Naive Bayes is much faster than KNN. Thus, it could be used for prediction in real time.

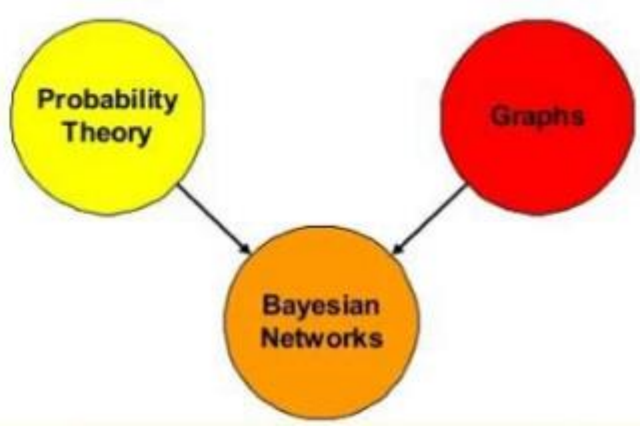
Naïve Bayes over Decision Tree

- Decision tree has a tendency to be over fitted with the training data.
- Decision tree are not generally suitable for application like diagnosis of Cancer. As Cancer doesn't occur in the population in large numbers, it may get pruned out more likely by decision tree.

Bayesian network

Bayesian Network and Its Uses:

- Common problems in real life :
  - Very complex in nature
  - Uncertainty



What is Bayesian Network ?

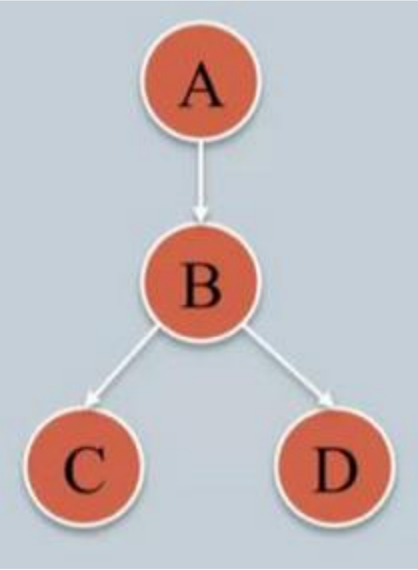
- A Bayesian network is a graphical model for depicting probabilistic relationships among a set of variables.

- It is represented in the form of Directed Acyclic Graphs (DAG).

- It is also called a Bayes Network,

Belief Network, Decision Network,

or Bayesian Model.



Bayesian Network constitutes of :

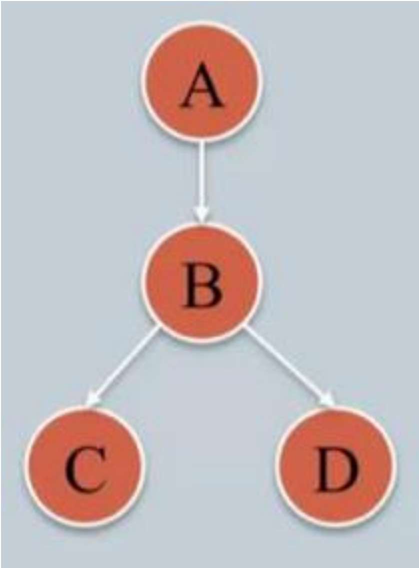
Directed Acyclic Graph (DAG): uses conditional probability for independent nodes and each of the dependent nodes is associated with a conditional probability table (CPT).

A	P(A)
false	0.6
true	0.4

Conditional Probability of A

A	B	P(B|A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

CPT for node B



B	D	P(D|B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

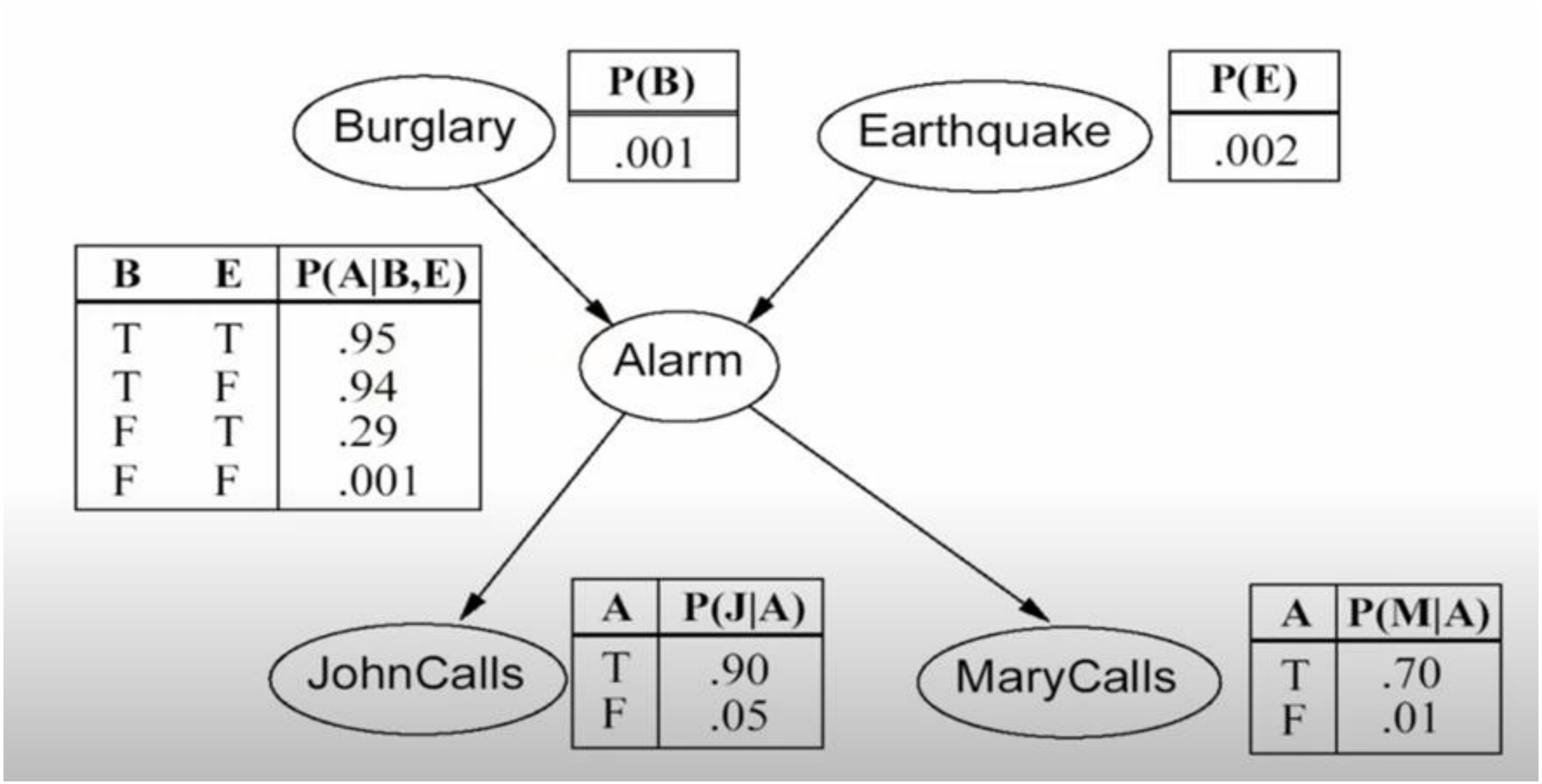
CPT for node D

B	C	P(C|B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

CPT for node C

- Bayesian Network encodes the conditional independence relationships between the variables in the graph structure.
- Provides a compact representation of the joint probability distribution among the variables.
- A problem domain is modeled by a list of variables $X_1,\ldots,X_n$
- Knowledge about the problem domain is represented by a joint probability $P(X_1,\ldots,X_n)$.
- Directed links represents casual direct influences.
- Each dependent node has a conditional probability table quantifying the effects from the parents.
- No directed cycles

Bayesian Network : An Example



P(B)
.001

P(E)
.002

B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

Bayesian Network : An Example

- You have a new burglar alarm installed at home.
- It is fairly reliable at detecting burglary, but also sometimes responds to minor earthquakes.
- You have two neighbors, John and Merry , who promised to call you at work when they hear the alarm.
- John always calls when he hears the alarm, but sometimes confuses telephone ringing with the alarm and calls too.
- Merry likes loud music and sometimes misses the alarm.
- Given the evidence of who has or has not called, we would like to estimate the probability of a burglary.

Bayesian Network : An Example

Each individual event has a probability such as, in this example:

Probability of burglary $P(b) = 0.001$

Probability of no burglary $P(\neg b) = 1 - 0.001 = 0.999$

Probability of earthquake $P(e) = 0.002$

Probability of no earthquake $= 1 - 0.002 = 0.998$

Probability of John calls given there is an alarm $P(j \mid a) = 0.90$

Probability of John not calls given there is an alarm $P(\neg j \mid a) = 1 - 0.90 = 0.10$

Probability of John calls given there is no alarm $P(j \mid \neg a) = 0.05$

Probability of John not calls given there is no alarm $P(\neg j \mid \neg a) = 1 -0.05 = 0.95$

Bayesian Network : An Example

Probability of Mary calls given there is an alarm $P(m \mid a) = 0.70$

Probability of Mary not calls given there is an alarm $P(\neg m \mid a) = 1 - 0.70 = 0.30$

Probability of Mary calls given there is no alarm $P(m \mid \neg a) = 0.01$

Probability of Mary not calls given there is no alarm $P(\neg m \mid \neg a) = 1 - 0.01 = 0.99$

Bayesian Network : An Example

Each node having conditional dependence with one or more events is associated with a Conditional Probability Table (CPT) such as, in our example:

The probability of alarm given different condition of burglary, earthquake, no burglary and no earthquake is given by its CPT.

Probability of alarm given burglary and earthquake $P(a \mid b,e) = 0.95$

Probability of alarm given no burglary and earthquake $P(a \mid \neg b,e) = 0.29$

Probability of alarm given burglary and no earthquake $P(a \mid b,\neg e) = 0.94$

Probability of alarm given no burglary and no earthquake $P(a \mid \neg b,\neg e) = 0.001$

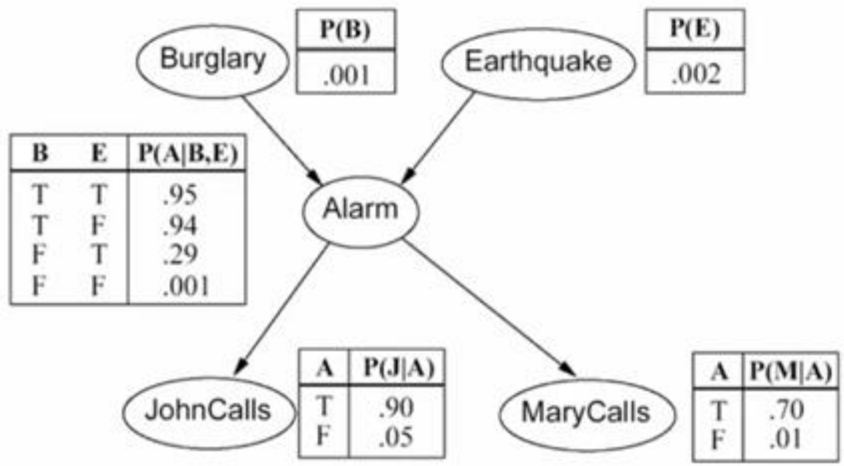
Bayesian Network : An Example

1. What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?

So, in this case, we have been given all the events i.e. burglary, earthquake, alarm, John and Merry. Considering all the events, we need to calculate the joint probability distribution i.e. $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$.

Bayesian Network : An Example

1. What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?



Solution:

$$P(j \wedge m \wedge a \wedge \neg b \wedge \neg e) = P(j \mid a)\, P(m \mid a)\, P(a \mid \neg b, \neg e)\, P(\neg b)\, P(\neg e)$$

$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998$$

$$= 0.00062$$

Bayesian Network : An Example

2. What is the probability that John call?

In this case, we have been given only one event that is John has called. We need to consider the Bayesian Network and solve this case.

Probability of John calling depends on alarm. So there are two possible cases :

A) John calls given there is an alarm, $P(j \mid a)P(a)$

B) John calls given there is no alarm, $P(j \mid \neg a)P(\neg a)$

So, $P(j) = P(j \mid a)P(a) + P(j \mid \neg a)P(\neg a)$

Bayesian Network : An Example

Now, we can get the value of P(a) from the given conditional probability table of alarm.

Probability of alarm and no alarm depends on burglary and earthquake. But we don't know anything about whether burglary and earthquake happened or not. Hence we need to consider all 4 possible cases for alarm and no alarm :

There is an alarm given : there is a burglary and there is an earthquake $P(a \mid b,e) = 0.95$

there is a burglary and there is no earthquake $P(a \mid b,\neg e) = 0.94$

there is no burglary and there is an earthquake $P(a \mid \neg b,e) = 0.29$

there is no burglary and there is no earthquake $P(a \mid \neg b,\neg e) = 0.001$

There is no alarm given : there is a burglary and there is an earthquake $P(\neg a \mid b,e) = 1 - 0.95 = 0.05$

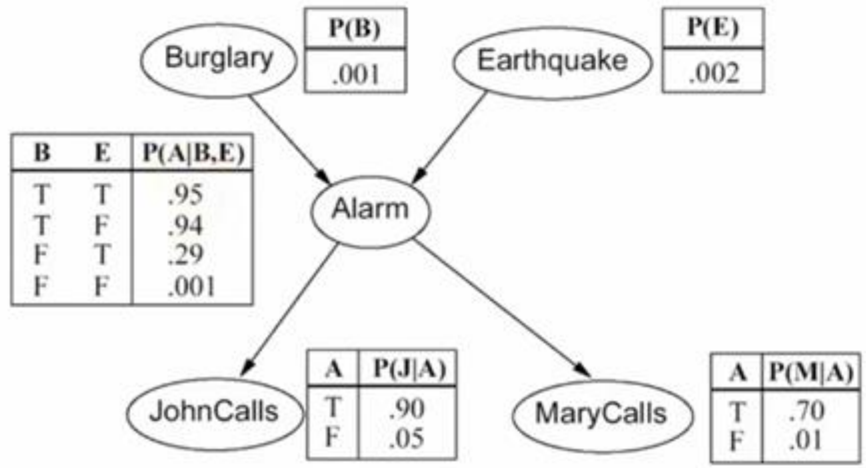
there is a burglary and there is no earthquake $P(\neg a \mid b,\neg e) = 1 - 0.94 = 0.06$

there is no burglary and there is an earthquake $P(\neg a \mid \neg b,e) = 1 - 0.29 = 0.71$

there is no burglary and there is no earthquake $P(P(\neg a \mid \neg b,\neg e) = 1-0.001 = 0.999$

Bayesian Network : An Example

2. What is the probability that John call?



P(B)
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B	E	P(A|B,E)
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T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

Solution:

$P(j) = P(j \mid a)\, P(a) + P(j \mid \neg a)\, P(\neg a)$

$= P(j|a)\{P(a|b,e)*P(b,e)+P(a|\neg b,e)*P(\neg b,e)+P(a|b,\neg e)*P(b,\neg e)+P(a|\neg b,\neg e)*P(\neg b,\neg e)\}$

$+ P(j|\neg a)\{P(\neg a|b,e)*P(b,e)+P(\neg a|\neg b,e)* P(\neg b,e)+P(\neg a|b,\neg e)* P(b,\neg e)+P(\neg a|\neg b, \neg e)*$

$P(\neg b, \neg e)\}$

$= 0.90 * 0.00252 + 0.05 * 0.9974 = 0.0521$

Bayesian network can also be used to represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases.

Typical Use of Bayesian networks

- To model and explain a domain.
- To update beliefs about states of certain variables when some other variables were observed, i.e., computing conditional probability distributions, e.g., $P(X_{23}|X_{17} = yes, X_{54} = no)$).
- To find most probable configurations of variables
- To support decision making under uncertainty
- To find good strategies for solving tasks in a domain with uncertainty.